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European Patent Application No. 22892488.2
Derived from PCT Application No. PCT/JP2022/038211
In the name of SUMITOMO ELECTRIC INDUSTRIES, LTD.
Our Ref: RPS/OJR/P258288EP00

In response to the communication pursuant to Rules 70(2) and 70a(2) EPC, dated 11 February 2025, I hereby confirm the applicant's desire to proceed further with this application.

The applicant would also like to take this opportunity to respond to the objections raised in the Search Opinion contained in the extended European search report of 23 January 2025.

Clarity (Article 84 EPC)

The Examiner considers that claim 3 is unclear because an intersection relates to a single point, not a line. However, the applicant respectfully disagrees. Claim 3 specifies "wherein a line of intersection of the connection surface and the cross-section including the rotation axis is a straight line". This effectively means that the connection surface is flat (see Fig.4) as opposed to curved (see Fig.5). The cross-section including the rotation axis is a plane, i.e. a two-dimensional surface. The connection surface is at 90° to this plane. Accordingly, the connection surface and cross-section intersect along one dimension. Hence, claim 3 refers to a line rather than to a single point.

Accordingly, it is submitted that the requirements of Article 84 EPC are fulfilled.

Novelty (Articles 52(1) and 54 EPC)

D1:

Claim 1 requires that the contactor is made of polycrystalline diamond including a plurality of diamond particles. As discussed in paragraph [0025] of the description, the polycrystalline diamond of the present invention is polycrystalline diamond in which diamond particles are directly bonded to each other, i.e. without a bonding / binder phase (often referred to as "BL-PCD"). In contrast, D1 uses sintered diamond that includes a metal bond (see, for example, paragraph [0034] of D1). Accordingly, claim 1 is novel over D1.

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D2:

D2 discloses a wheel equipped with a hard substrate, such as tungsten carbide, and a diamond coating on its surface (see, for example, claim 1 and claim 5 of D2). D2 does not disclose the use of polycrystalline diamond (BL-PCD) as per claim 1. Accordingly, claim 1 is novel over D2.

D3:

The wheel material in D3 consists of diamond materials that include metal bonds or resin bonds (see, for example, the section entitled "1. Instruction"). D3 does not disclose the use of polycrystalline diamond (BL-PCD) as per claim 1. In addition, D3 features a wheel diameter with a radius of 1 cm. This is significantly larger in size than the “length from the rotation axis to the outer end portion” of claim 1, which is “0.5 mm to 5 mm”. Accordingly, claim 1 is novel over D3.

D4:

D4 does not disclose the use of BL-PCD, and only PCD is mentioned in paragraph [0025] of D4. Accordingly, claim 1 is novel over D4.

D5:

In Claim 1 of D5, the range for the average diamond particle size is specified as "50 to 2500 nm," which is considerably broader than the range of "10 to 300 nm" in Claim 1 of the present invention. In addition, D5 does not disclose a contactor. Accordingly, claim 1 is novel over D5.

Inventive Step (Articles 52(1) and 56 EPC)

Claim 1

The closest prior art could be considered to be document D1.

As discussed above, one distinction between the contactor of claim 1 and the disclosure of D1 is the D1 does not disclose that the contactor is made of polycrystalline diamond including a plurality of diamond particles.

The technical effect conferred by this difference, in combination with the claimed average particle diameter of 10 nm to 300 nm, is that the contactor exhibits isotropic hardness and abrasion resistance with respect to the orientation. Accordingly, the contactor can be used for evaluating the abrasion

characteristics in the minute region of the polycrystalline diamond. This is discussed, *inter alia*, in paragraphs [0011], [0025], [0028] and [0029] of the description.

Starting from the closest prior art according to D1 the objective technical problem underlying the present invention is how to enable the contactor to evaluate the abrasion characteristics in the minute region of polycrystalline diamond.

D1 provides no suggestion that the above problem can be solved by the combination of features set out in claim 1. As discussed above, D1 uses sintered diamond that includes a metal binder. There is no suggestion in D1 to replace such diamond with polycrystalline diamond including a plurality of diamond particles. This is also no suggestion that such use of polycrystalline diamond in combination with the average particle diameter of 10 nm to 300 nm would enable the contactor to evaluate abrasion characteristics in the minute region. The same arguments apply also to D2-D5, since these documents also do not disclose the use of polycrystalline diamond in a contactor. In fact, paragraph [0004] of D2 appears to teach away from the present invention since it discusses issues associated with forming the entire wheel from polycrystalline diamond. Thus, D1 alone, or in combination with any of D2-D5, provides no pointer or incentive to the present invention.

For the above reasons it is submitted that the solution to the objective technical problem underlying the present invention is not obvious the skilled person. Therefore, claim 1 involves an inventive step for at least this reason.

Claim 5

It is noted that claim 5 involves and inventive step, at least by virtue of its dependency on claim 1. It is submitted that claim 5 involves an inventive step for at least the following additional reasons. Neither D4 nor D5 discloses or suggests the use of a contactor to make contact with single crystal diamonds (D4 is for glass scribing applications), and the effect of evaluating the wear characteristics in the microscopic area of single crystal diamonds as described in the present invention (paragraph 0011) is not something that can be easily conceived from the prior art. Furthermore, none of the cited references discloses a configuration in which a contactor made of "BL-PCD" is brought into contact with "single crystal diamond." Accordingly, it is submitted that claim 5 involves an inventive step for these additional reasons.

Miscellaneous

It is submitted that the dependent claims are also novel and inventive, at least by virtue of their dependencies.

Amendment of the description has been deferred until the wording of the claims have been finalised.

It is submitted that the claims are in order for allowance. In the event that the Examining Division disagrees to the extent that refusal of the application is contemplated, oral proceedings are requested.

Yours faithfully,

Dr Rohan P. SETNA
For and on behalf of BOULT WADE TENNANT LLP
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